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$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$$

$$\frac{\partial f}{\partial x}(x, y, z) = \frac{1}{2} \frac{1}{\sqrt{x^2 + y^2 + z^2}} 2x = \frac{x}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{\partial f}{\partial y}(x, y, z) = \frac{1}{2} \frac{1}{\sqrt{x^2 + y^2 + z^2}} 2y = \frac{y}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{\partial f}{\partial z}(x, y, z) = \frac{1}{2} \frac{1}{\sqrt{x^2 + y^2 + z^2}} 2z = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\nabla f(x, y, z) = \left(\frac{x}{\sqrt{x^2 + y^2 + z^2}}, \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$\nabla f(12, 3, 4) = \left(\frac{12}{13}, \frac{3}{13}, \frac{4}{13} \right)$$

References